

Cross-dataset transfer of a Boltz raw-Δz ΔΔG predictor

Experiment 05 — Tsuboyama → FireProt. ddG_with_Boltz project · Generated 2026-07-19 · raw-Δz features, MLP (5-seed ensemble) primary model.

A ΔΔG predictor trained on the entire **Tsuboyama** mega-scale folding dataset (12,359 mutations) transfers — with **no refitting** — to the independent **FireProt** dataset (3,205 mutations / 138 proteins, **zero protein overlap**): pooled Pearson **r = 0.648**, Spearman **ρ = 0.659**, per-protein **median r = 0.65**. The Boltz raw-Δz signal is not an artifact of one dataset. Its ceiling is **magnitude, not ranking**: predictions span only ~40% of the true ΔΔG spread (fit slope 0.27), so the model ranks and triages well but under-predicts absolute effect sizes on out-of-range mutations.

1. Objective

Every generalization test so far (experiments 01, 06) splits *within* the Tsuboyama corpus. Such holdouts can still ride on quirks shared by a single data source — Tsuboyama is one high-throughput folding assay on mostly small/designed domains. The decisive question for a usable predictor is whether the signal carries to an **independently curated** dataset: different proteins, a different assay, and a different label provenance. We test transfer to **FireProt** (literature-derived stability measurements, ≤500 aa).

2. Data & methods

- Train:** all 12,359 Tsuboyama mutations (`tsuboyama_bench_fast/rawz_features.parquet`), **256 raw-Δz features** (mutant–WT difference of the Boltz-2 pair track `z: zdiag_* 128 + zpool_* 128`).
- Test:** all 3,205 FireProt mutations / 138 proteins (`fireprot_le500/features_summary.parquet`), the *same* 256 features.
- Model:** the benchmark pipeline `SimpleImputer(median) → StandardScaler → estimator`, fit once on Tsuboyama and applied unchanged to FireProt. Primary estimator: a **5-seed MLP ensemble** (project default since exp 06); HistGradientBoosting (HGB) reported alongside.
- Entry point:** `python -m ddg.evaluation.transfer --train ... --test ... --model mlp`.
- Independence:** the train and test protein sets have **zero wt_id overlap**. FireProt's ΔΔG range [-13.7, 12.0] is much wider than Tsuboyama's [-2.7, 5.7].
- Sign convention:** FireProt (`ddG`) and Tsuboyama (`ddg`) share the same sign (positive = destabilizing; both ~75% positive), so predictions are used as-is (no flip).

3. Results

Metric	MLP (primary)	HGB
Pooled Pearson r	0.648	0.616
Pooled Spearman ρ	0.659	0.631
Pooled RMSE	1.28	1.34
Pooled MAE	0.84	0.89
Per-protein r — mean	0.370	0.419
Per-protein r — median	0.647	—
Proteins scored	114 / 138	114 / 138

Table 1. Pooled and per-protein transfer metrics (n = 3,205 mutations / 138 proteins). ΔΔG in kcal/mol.

MLP and HGB are within noise of each other — the same "representation, not model" result as experiment 06. 60% of proteins score $r > 0.5$ and 41% score $r > 0.7$; the per-protein *mean* (0.37) sits well below the *median* (0.65) because a handful of proteins transfer poorly (Fig. 2).

Transfer Tsuboyama $\rightarrow$ FireProt ≤ 500
training range [-1.0, 4.0] kcal/mol shaded

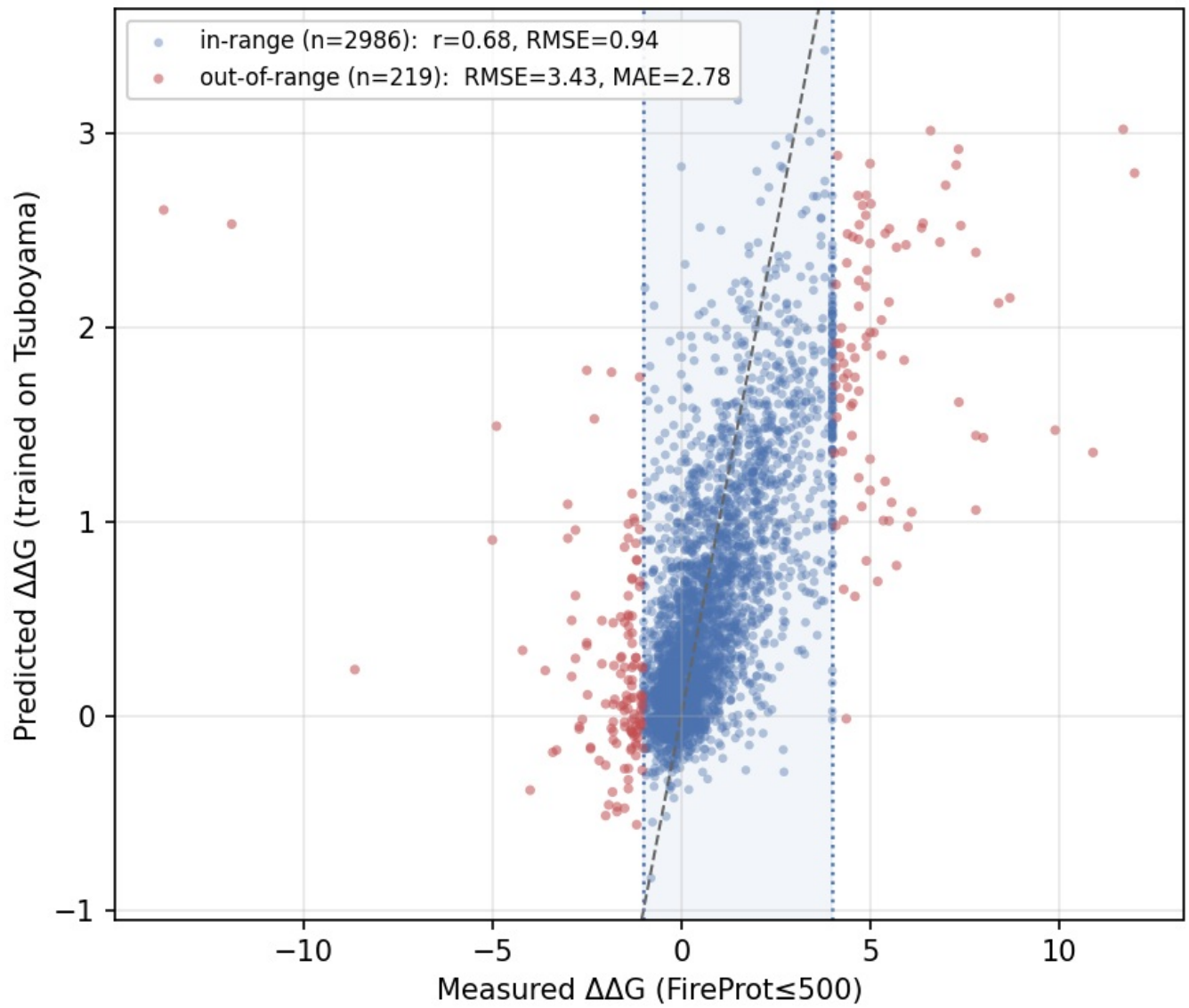


Figure 1. Predicted (trained on Tsuboyama) vs measured (FireProt) $\Delta\Delta G$ for all 3,205 mutations. Dashed line is $y = x$. The cloud is strongly correlated but far flatter than the diagonal: predicted $\Delta\Delta G$ concentrates in $\sim[0, 3]$ while measured values span $[-13.7, 12.0]$ — the magnitude compression quantified in §4.

Per-protein transfer (n=114 scored, of 138)

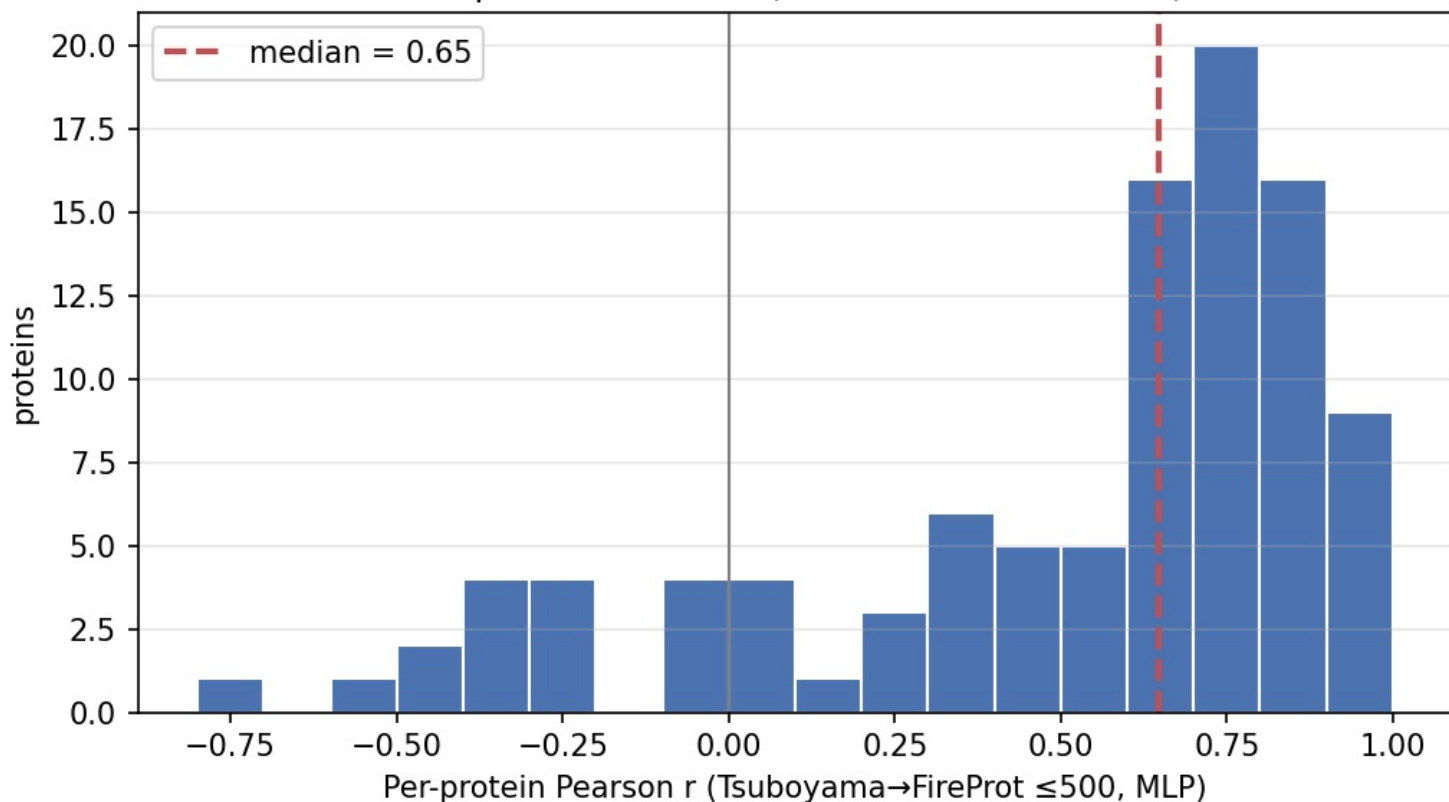


Figure 2. Distribution of per-protein Pearson r (114 of 138 proteins scored; the rest have <2 mutations or constant $\Delta\Delta G$). Concentrated at 0.6–1.0 (median 0.65); the mean 0.37 is pulled down by a few poorly-transferring proteins.

Protein	n	Pearson r	Spearman ρ	RMSE
P08463	14	0.895	0.664	0.853
P01053	49	0.877	0.900	0.896
P02417	26	0.864	0.875	0.996
P0AA04	12	0.847	0.792	0.930
P00648	114	0.835	0.740	1.663
P03034	21	0.832	0.702	1.167
... 103 proteins between ...				
P08921	8	-0.958	-0.905	2.899
3PG0	10	-0.447	-0.148	1.061
Q08012	8	-0.404	-0.012	1.605
P11540	9	-0.359	-0.452	1.006
P37001	19	-0.284	-0.319	0.658

Table 2. Best- and worst-transferring proteins (≥ 8 mutations each).

4. The ceiling: magnitude, not ranking

The predictor *ranks* mutations well ($\rho = 0.66$) but severely **under-predicts magnitude**. The predicted-vs-measured fit slope is **0.27**, and predicted $\Delta\Delta G$ has SD 0.66 against a measured SD of 1.58 — only ~42% of the true spread. FireProt's wide range makes this starker than any within-Tsuboyama split: the model regresses toward the mean on the destabilizing/stabilizing tails it never saw in training. This is the same regression-to-the-mean weakness isolated in experiment 02 (extrapolation) and confirmed model-independent in 06 — a property of the features/objective, not of the estimator. **Practical consequence:** use for ranking, prioritization, and triage — not for absolute $\Delta\Delta G$ on mutations outside the training range.

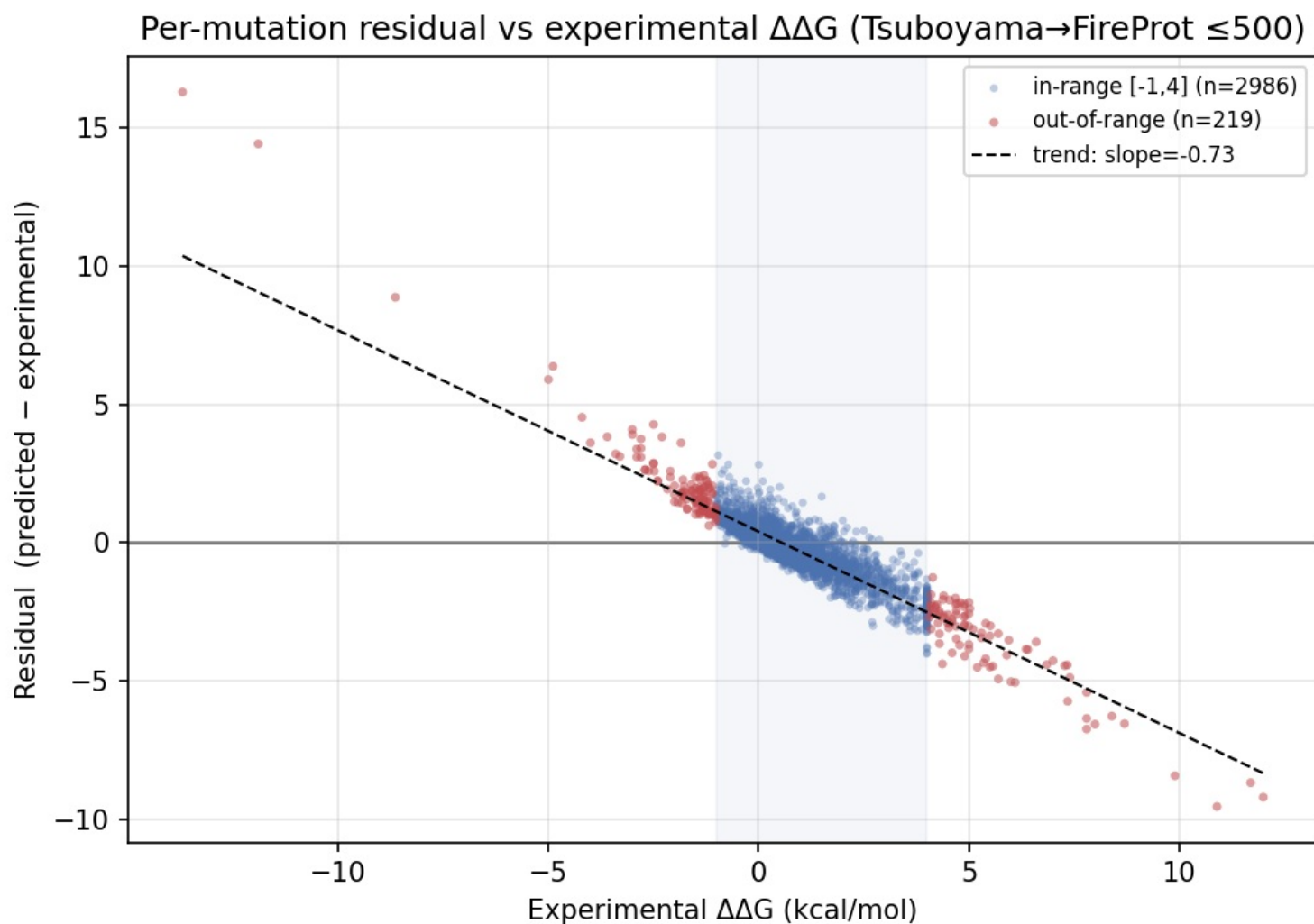


Figure 3. Per-mutation residual (predicted – experimental) vs experimental $\Delta\Delta G$. The residual trends downward with a slope of ≈ -0.73 (dashed line): the model over-predicts strongly stabilizing mutations and under-predicts strongly destabilizing ones, crossing zero only near the dense training centre. In-range residuals cluster near zero (mean $|\text{residual}|$ 0.69 kcal/mol); out-of-range residuals fan out (mean $|\text{residual}|$ 2.8).

5. Where the error lives: training coverage of $\Delta\Delta G$

Splitting the test by the Tsuboyama training range ($[-1, 4]$ kcal/mol, its central $\sim 98\%$) makes the ceiling precise. **In-range** (n=2986) the model is genuinely good: Pearson **0.68**, RMSE **0.94**. **Out-of-range** (n=219) the error explodes: RMSE **3.43**, and the per-tail correlation collapses — the predictions cannot reach $\Delta\Delta G$ values absent from training.

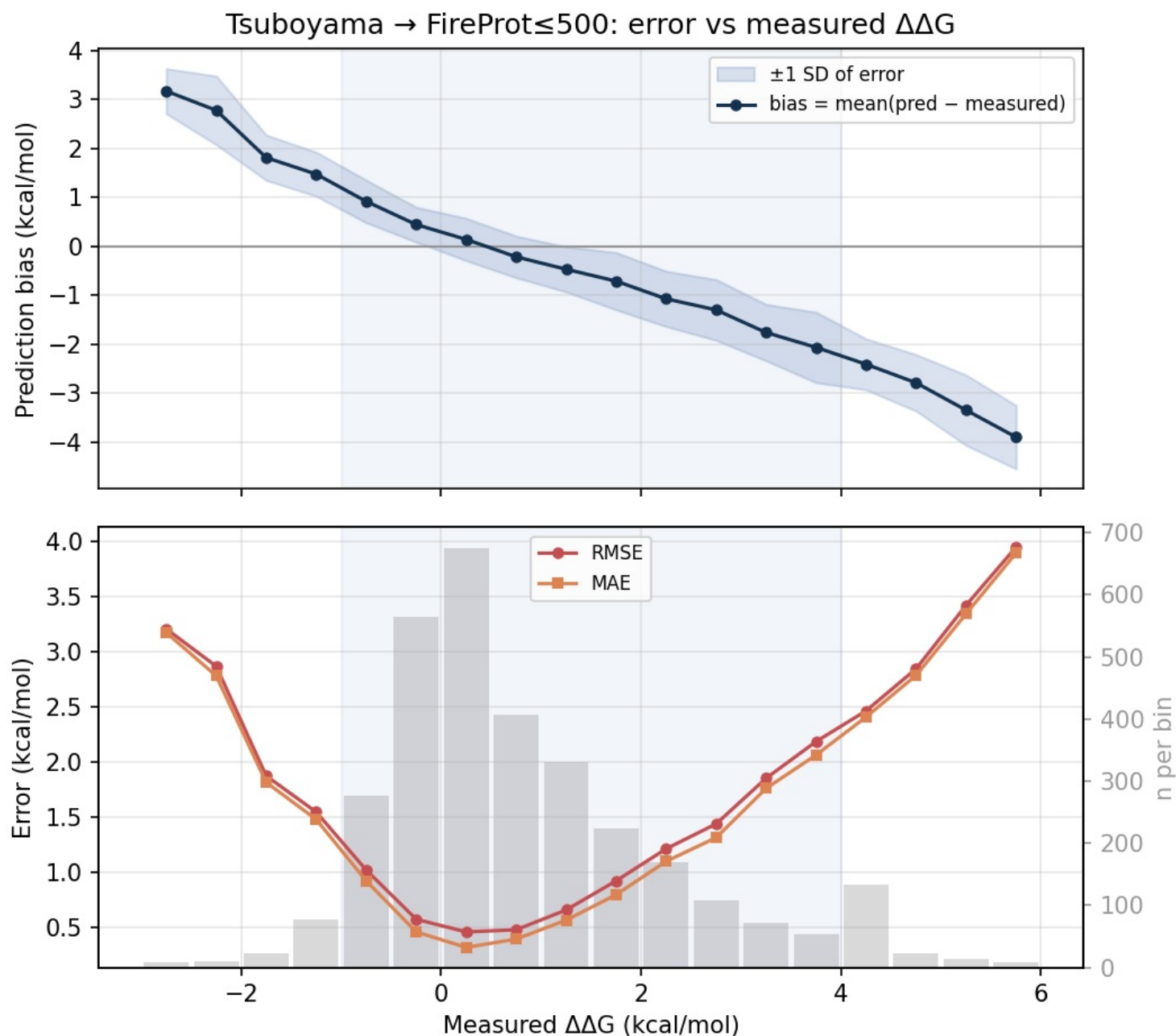


Figure 4. Prediction error vs measured $\Delta\Delta G$. A regression-to-mean bias (over-predicts low $\Delta\Delta G$, under-predicts high), with error minimized in the dense centre and rising toward both tails; the $\pm$ SD band is flat, so tail error is systematic bias.

The cause is **training density**, not FireProt itself: relating per-bin error to how densely Tsuboyama sampled each $\Delta\Delta G$ value, error is almost perfectly anti-correlated with density (Spearman $\rho = -0.96$). Accuracy at a given $\Delta\Delta G$ is set by how much training data covered it — a coverage effect, identical across model families (cf. experiments 02 and 06).

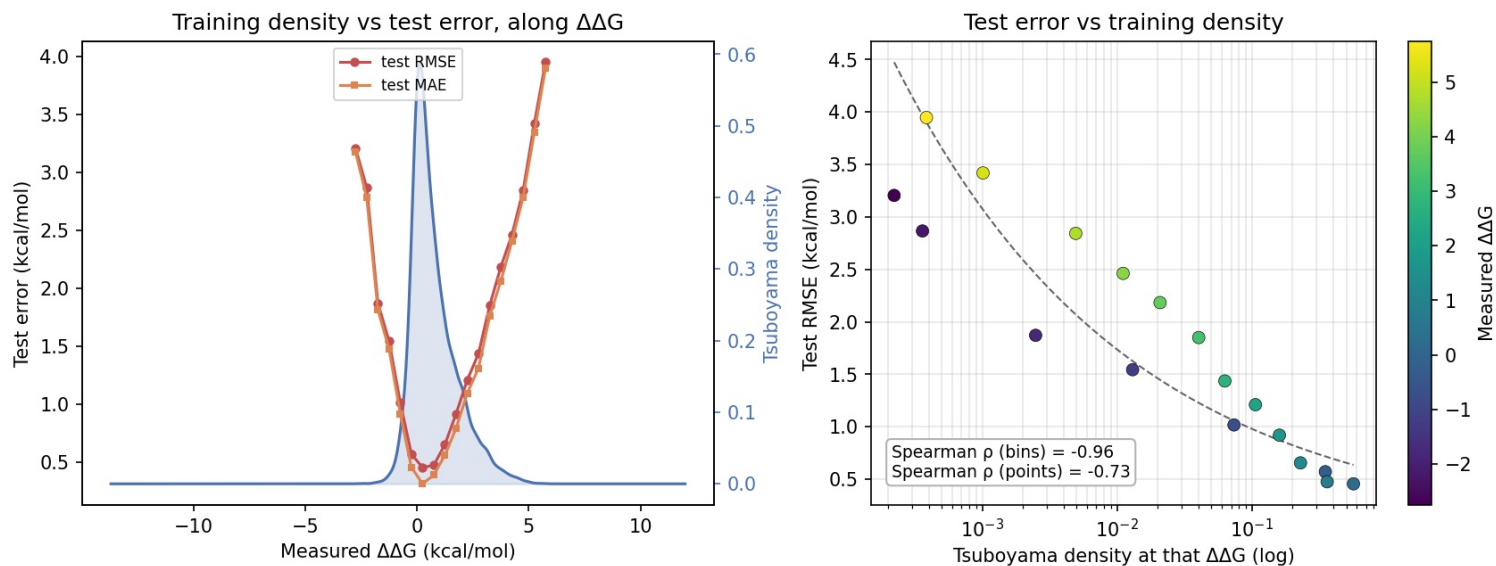


Figure 5. Test error vs Tsuboyama training density in $\Delta\Delta G$ space. Left: density and error are mirror images along $\Delta\Delta G$. Right: error falls monotonically with training density.

6. Context

The pooled $r \approx 0.65$ matches the published state of the art for this transfer: ThermoMPNN and the AFToolkit framework report ~ 0.65 Pearson transferring to FireProt from the same Megascale/Tsuboyama training data using AlphaFold2 / graph-neural-network backbones. The Boltz-2 raw- Δz pipeline reaches the same level with a simple downstream regressor.